

Schematic Notes

USB-C Connector:

- The USB-C connector is used as the 5V input source for the charging circuit.
- 5.1kΩ resistors are connected to the CC1 and CC2 pins for proper USB-C power detection.
- Only the VBUS and GND lines are used since the USB connection is intended mainly for charging.

TP4056 Charging Circuit

- The TP4056 IC is used for safe charging of the single-cell LiPo battery.
- A 2.0 kΩ resistor is connected to the PROG pin to set the charging current to approximately 580mA.
- 10μF capacitors are added near the input and battery output for voltage stabilization and noise reduction during charging.

AMS1117-3.3 Voltage Regulator

- The AMS1117-3.3 regulator is used to convert the battery voltage into a stable 3.3V supply.
- 10μF capacitors are connected at the input and output for voltage stabilization.
- An additional 100nF capacitor is added at the output for high-frequency noise filtering.
- The regulated 3.3V output powers both the ESP32 and MPU6050 sensor.

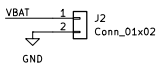
ESP32 Module

- The ESP32 acts as the main microcontroller for sensor processing and embedded control.
- A 10kΩ pull-up resistor is connected to the EN pin to keep the ESP32 enabled during operation.
- 10μF and capacitor is added near the power pins for stable operation and noise reduction.
- The ESP32 communicates with the MPU6050 using the I2C protocol.

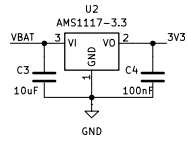
MPU6050 Sensor

- The sensor communicates with the ESP32 through the SDA and SCL I2C communication lines.
- 4.7kΩ pull-up resistors are used on the SDA and SCL lines for stable I2C communication.
- A 100nF decoupling capacitor is connected near the VCC pin to reduce noise.
- Additional capacitors connected to the REGOUT and CPOUT pins are added according to the MPU6050 datasheet recommendations.

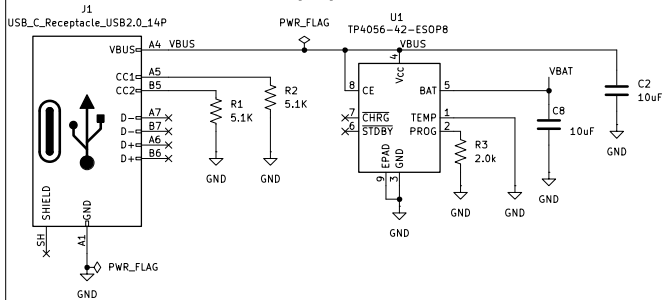
Lipo Battery



LDO



Charging Circuit



ESP32 Sensor Processing Unit

